UdaPlay - Project Writeup

1. Overview

UdaPlay is an AI-powered research agent designed to answer natural-language questions about video games. The system combines a local retrieval model (RAG), an LLM-based evaluator, and a web search fallback using Tavily. The project is implemented in two notebooks:

- Udaplay_01_starter_project.ipynb
- Udaplay_02_starter_project.ipynb

This writeup describes how each component satisfies the project rubric.

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2. Vector Database and RAG Pipeline

2.1 Dataset Preparation

In Notebook 1, all JSON game files provided in `starter/games/` were loaded and processed. Each file was converted into an indexed document containing:

- Name
- Platform
- Year of Release
- Description
- All additional metadata

The text content was prepared using a concise formatted string combining the key fields.

2.2 Vector Database Construction

A persistent ChromaDB client was initialized at the path `chromadb`.
A collection named `udaplay` was created using `OpenAIEmbeddingFunction` with the Vocareum OpenAI endpoint.

All processed documents were embedded and added to this persistent collection.

2.3 Semantic Search Demonstration

The notebook includes a dedicated section demonstrating semantic search functionality. Several sample queries were run, and the top results were displayed with:

- Document text
- Metadata
- Distances

This confirms that the RAG pipeline is functioning correctly.

This completes the requirements for preparing, processing, and querying a local vector database as described in the rubric.

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3. Agent Tools

Notebook 2 implements three required tools using the project's `@tool` decorator.

3.1 retrieve_game

This tool performs semantic search over the Chroma vector database. It returns a list of structured dictionaries containing:

- Name
- Platform
- Year of Release
- Genre
- Publisher
- Description
- Vector distance

This satisfies the requirement for an internal retrieval tool.

3.2 evaluate_retrieval

This tool uses the LLM (configured to use the Vocareum endpoint) to assess whether the internal documents retrieved are sufficient to answer the user's question. It returns a JSON-compatible structure with fields:

- useful: Boolean
- description: Explanation of the decision

This serves as the LLM-as-judge component.

3.3 game_web_search

This tool uses the Tavily API to perform a web search when internal information is insufficient. It returns the Tavily response, which includes:

- An AI-generated summary (answer)
- A list of web results with titles, URLs, and snippets

This satisfies the requirement for a web search fallback tool.

All three tools are integrated into the agent workflow.

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4. Stateful Agent

A custom class `UdaPlayAgent` was implemented to manage conversation flow using a clear state machine represented by an enum. The states are:

- RETRIEVE
- EVALUATE
- WEB_SEARCH
- ANSWER

4.1 Workflow

1. Retrieve
   The agent first calls `retrieve_game` to obtain internal documents.

2. Evaluate
   The agent calls `evaluate_retrieval` to determine whether internal documents are sufficient.

3. Web Search (conditional)
   If the evaluation indicates insufficiency, `game_web_search` is invoked.

4. Answer
   The agent synthesizes a final answer using the main LLM.
   The answer includes clearly formatted source information.
   Internal answers end with "Source: Local database."
   Web answers include specific URLs.

4.2 State and Memory

The agent maintains a list of past interactions in `self.history`, allowing it to process multiple queries within the same session. This satisfies the rubric requirement for a stateful agent capable of handling multiple questions.

4.3 Output Quality

The agent produces clear, structured answers that cite their sources.
This satisfies the rubric requirement for well-cited final answers.

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5. Demonstration of Agent Performance

Three sample queries were executed in the final notebook as required.

5.1 Query: When were Pokémon Gold and Silver released

- Internal RAG was insufficient according to the evaluator.
- The agent performed a web search.
- Final answer included accurate dates and multiple web citations.
- Web search usage was correctly indicated in the debug output.

5.2 Query: Which one was the first 3D platformer Mario game

- Internal RAG was sufficient.
- The agent correctly identified Super Mario 64 based solely on the local database.
- The final answer included the required citation: "Source: Local database."

5.3 Query: Was Mortal Kombat X released for PlayStation 5

- Internal RAG did not contain relevant information.
- The agent used Tavily, retrieved accurate information, and cited relevant URLs.
- The answer distinguished between native PS5 release and backward compatibility.

5.4 Debug Information

Each example prints:

- The evaluation report
- Retrieved internal documents
- Whether web search was invoked

This fulfills the rubric requirement for demonstrating agent reasoning and tool usage.

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6. Conclusion

All rubric requirements are fully met:

- A complete RAG pipeline with persistent vector storage
- Three functional tools integrated into an agent workflow
- A stateful agent with a clear state machine structure
- Correct fallback logic from internal knowledge to web search
- Well-structured, well-cited answers
- Demonstration on at least three example queries with tool usage explained

The UdaPlay agent successfully delivers a full hybrid retrieval system using both local knowledge and real-time web search, consistent with the project specifications.

Output

Screenshots

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================================================================================
QUESTION: When were Pokémon Gold and Silver released?

ANSWER:

Pokémon Gold and Silver were released in Japan on November 21, 1999, and in North America on October 15, 2000. These games were the first i

Sources:
- Web 1: https://www.pokemon.com/us/pokemon-video-games/pokemon-gold-version-and-pokemon-silver-version
- Web 3: https://en.wikipedia.org/wiki/Pok%C3%A9mon_Gold_and_Silver

--- Debug info ---
Evaluation: {'useful': False, 'description': 'The retrieved documents do not provide the release date for Pokémon Gold and Silver.'}
Num internal docs: 5
Web search used: True

================================================================================
QUESTION: Which one was the first 3D platformer Mario game?

ANSWER:

The first 3D platformer Mario game is **Super Mario 64**, released in 1996 for the Nintendo 64. This game was groundbreaking for its time,

Sources:
- Local database

--- Debug info ---
Evaluation: {'useful': True, 'description': 'The retrieved documents include information about Super Mario 64, which is the first 3D platfo
Num internal docs: 5
Web search used: False

================================================================================
QUESTION: Was Mortal Kombat X released for PlayStation 5?

ANSWER:

Mortal Kombat X was not released specifically for the PlayStation 5. It was originally launched for PlayStation 4, Xbox One, and PC in Apri

Sources:
- Local database
- [Web 2](https://www.playstation.com/en-us/games/mortal-kombat-x_msm_moved/)
- [Web 4](https://mortalkombat.fandom.com/wiki/Mortal_Kombat_X)

--- Debug info ---
Evaluation: {'useful': False, 'description': 'The retrieved documents do not contain any information about Mortal Kombat X or its release o
Num internal docs: 5
```

```
--- Debug info ---
Evaluation: {'useful': False, 'description': 'The retrieved documents do not contain any information about Mortal Kombat X or its release o
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Web search used: True
```

Text

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QUESTION: When were Pokémon Gold and Silver released?

ANSWER:

Pokémon Gold and Silver were released in Japan on November 21, 1999, and in North America on October 15, 2000. These games were the first in the Pokémon series designed specifically for the Game Boy Color.

Sources:
- Web 1:
https://www.pokemon.com/us/pokemon-video-games/pokemon-gold-version-and-pokemon-silver-version
- Web 3: https://en.wikipedia.org/wiki/Pok%C3%A9mon_Gold_and_Silver

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Num internal docs: 5
Web search used: True

==
QUESTION: Which one was the first 3D platformer Mario game?

ANSWER:

The first 3D platformer Mario game is **Super Mario 64**, released in 1996 for the Nintendo 64. This game was groundbreaking for its time, setting new standards for the platforming genre and featuring Mario's quest to rescue Princess Peach.

Sources:
- Local database

--- Debug info ---
Evaluation: {'useful': True, 'description': 'The retrieved documents include information about Super Mario 64, which is the first 3D platformer Mario game.'}
Num internal docs: 5
Web search used: False

==
QUESTION: Was Mortal Kombat X released for PlayStation 5?

ANSWER:

Mortal Kombat X was not released specifically for the PlayStation 5. It was originally launched for PlayStation 4, Xbox One, and PC in April 2015. While it can be played on PlayStation 5 through backward compatibility, there was no dedicated version or release for the PS5 itself.

Sources:
- Local database

- [Web 2](https://www.playstation.com/en-us/games/mortal-kombat-x_msm_moved/)
- [Web 4](https://mortalkombat.fandom.com/wiki/Mortal_Kombat_X)

--- Debug info ---
Evaluation: {'useful': False, 'description': 'The retrieved documents do not contain any information about Mortal Kombat X or its release on PlayStation 5.'}
Num internal docs: 5
Web search used: True